

Nicky J. van den Berg

✉ nicky.vandenberg@maastrichtuniversity.nl ◇ [in linkedin.com/in/nickyvandenberg](https://www.linkedin.com/in/nickyvandenberg) ◇ [🌐 nickyjvandenberg.com](https://nickyjvandenberg.com)

PROFILE

I completed my Ph.D. in (Applied) Mathematics, specializing in geometric image processing. My motivation is fueled by the desire to make a positive impact on society. Therefore, my personal goal is to develop image processing techniques that improve quality of life.

Previously, I developed techniques for processing retinal images, such as tracking and automatic grouping of vascular trees. I am eager to leverage my experience in image analysis for the automatic analysis of ophthalmological images and data.

EXPERIENCE

Postdoctoral Researcher, University Eye Clinic Maastricht 03 - 2026 – present

Using my experience with image analysis to analyse the ophthalmological data from “De Maastricht Studie”.

Postdoctoral Researcher, Eindhoven University of Technology (TU/e) 03 - 2026 – present

Seconded to TU/e for two days per week to continue the research started during my PhD and to strengthen the collaboration between the University Eye Clinic and TU/e.

PhD Candidate, Eindhoven University of Technology (TU/e) 11 - 2021 – 11 - 2025

Title Thesis: *Geometric Tracking and Grouping of Complex Vasculatures in Medical Images*.

Supervision: Remco Duits, Olga Mula, Tos Berendschot

Financed by the NWO Talent Programme VICI 2020 Exact Sciences (Duits, Geometric Learning for Image Analysis, VI.C 202–031).

Visiting Researcher, Emory 10 - 2024 – 10 - 2024

Host: Lars Ruthotto

Financed by the EU REMODEL program (MSCA-SE 101131557).

Teaching Assistant, TU/e 09 - 2015 – 08 - 2021

Supervising tutorials for students from various programs (group size between 8 and 120 students) and correcting homework exercises.

Technical Author, Numworx 12 - 2019 – 06 - 2020

Developing mathematics exercises at a high school and college level for a digital mathematics platform.

Student Mentor, TU/e 09 - 2015 – 02 - 2017

Guiding freshmen in the transition to university.

EDUCATION

Eindhoven University of Technology (TU/e) 04 - 2018 – 07 - 2021

Master Industrial and Applied Mathematics cum laude

Master Thesis: *Improved Computational Schemes for Geodesic Vessel Tracking in Orientation Scores*

Supervision: Remco Duits, Bart M.N. Smets

University of Western Ontario (UWO) 02 - 2020 – 06 - 2020

Research Internship

Title Research Report: *The Blood Coagulation Process: A Mathematical Model*

Supervision: Mikko Karttunen, Barry Koren

Eindhoven University of Technology (TU/e) Bachelor Applied Mathematics with courses from Psychology & Technology Bachelor Thesis: <i>Disease Dynamics in the Framework of Interacting Particle Systems: The Position Dependent SIS-model</i> Supervision: Oliver Tse	09 - 2014 – 03 - 2018
Provinciale Secundaire School Voeren (PSS Voeren); High School Modern Languages - Science Extra-curricular courses in mathematics and science (physics and chemistry)	09 - 2008 – 06 - 2014

OTHER ACTIVITIES

Organizing Committee of SSVM 2027 , Conference Organizing the Scale Space and Variational Methods (SSVM) conference in Veldhoven, the Netherlands, in June 2027.	03 - 2026 – present
Founder and Board Member , Stichting Ada Alumni Founding a foundation for the Mathematics and Computer Science department alumni at TU/e and organizing a variety of activities for this audience. Functions: Secretary (04 - 2022 – 10 - 2022), Treasurer (10 - 2023 – 01 - 2026).	04 - 2022 – 01 - 2026
Workshop for high school teachers on AI , 4TU.NIRICT Presenting current image processing research (partially relying on) AI to mathematics high school teachers. Financed by 4TU.NIRICT to raise awareness in high schools on ICT challenges.	11 - 2025 – 11 - 2025
Organization of CASA outing , TU/e Organizing a day for the cluster consisting of a small activity, two museum visits, and lunch and dinner.	05 - 2025 – 06 - 2025
Organization of PhD activity , TU/e Organizing a dinner and evening activity for other PhD students.	11 - 2024 – 01 - 2025
Committee Member , E.P.A. Nexus (PhD association) Representation Committee (04 - 2023 – 11 - 2024) & Cash Control Committee (11 - 2023 – 11 - 2024)	04 - 2023 – 11 - 2024
Lil'Scientist Guiding primary school students through their first research project. This is a project of "De Jonge Akademie".	03 - 2024 – 06 - 2024
PROOF - PhD Development Sounding Board , TU/e Improving the experience and offer of PROOF-courses for PhD candidates.	09 - 2023 – 09 - 2023
Panel Discussion Research Day , TU/e On the impact of research.	06 - 2023 – 06 - 2023
Organization of CASA day , TU/e Organization of a day for the cluster where ongoing research is discussed, combined with a skill-session (e.g. workshop presenting skills, or a relevant lecture).	06 - 2022 – 05 - 2023
Curriculum Committee , TU/e Giving feedback on the newly developed curriculum for the bachelor Applied Mathematics.	02 - 2022 – 07 - 2022
Research Assessment , TU/e Part of the PhD candidate representation during interviews with the panel.	03 - 2022 – 03 - 2022
Student Member Program Committee Mathematics , TU/e Checking the quality of education (in co-determination). Representing the program committee in various general meetings, accreditations, and the ITK (quality assurance institutional assessment, see website NVAO).	09 - 2015 – 04 - 2021

Student Member Department Council , TU/e	01 - 2018 – 12 - 2019
Advising the department board while focusing on communication, well-being, and education.	
Secretary , Study association GEWIS	07 - 2017 – 06 - 2018
Responsible for the well-being of the association during the academic year 2017-2018.	
Responsible for the member administration, minutes of board and general meetings, and all mail.	
Various Committees , Study association GEWIS	09 - 2014 – 10 - 2021
Including developing a multi-annual vision, corporate identity, revision of the bylaws and internal regulations, etc.	
Activities in High School , PSS Voeren	09 - 2012 – 06 - 2014
Chair and member of the student council.	
Member of the school council.	

TEACHING

Tutor/teaching assistant for the following courses:

Numerical Linear Algebra	2024 – 2024
Applied Mathematics students	
Complex Analysis	2022 – 2024
Applied Mathematics students	
Calculus	2022 – 2024
Built Environment students	
Mathematics 1	2018 – 2021
Electrical Engineering and Industrial Engineering students	
Calculus	2017 – 2019
Applied Physics students	
Data Analytics for Engineers	2017 – 2018
Electrical Engineering, Applied Mathematics, and Psychology & Technology students	
Introduction to Modeling: From Problems to Numbers and Back	2016 – 2017
Psychology & Technology students	
Calculus	2015 – 2017
Mechanical Engineering, and Medical Sciences & Technology students	

SUPERVISION

Bachelor Project , Franciska Asma	02 - 2024 – 11 - 2024
Title Thesis: <i>Curvature Estimation and Adaptation for Chip Components Segmentation in SEM Images</i>	
Graduation Project , Leanne Vis	09 - 2023 – 07 - 2024
Title Thesis: <i>Automatic Edge-detection and Grouping of Possibly Overlapping Components in SEM Images of Silicon Wafers</i>	
Modeling Week Project	11 - 2023 – 11 - 2023
Title Report: <i>Geodesic Tracking of Blood Vessels in Wide-Field Retinal Images</i> (in $SO(3)$)	
Research Topic , Leanne Vis	03 - 2023 – 06 - 2023
Title Research Report: <i>Connected Components</i>	
Research Internship , Romain Dugast	04 - 2022 – 08 - 2022
Title Research Report: <i>Geometric Tracking in $SE(2)$ and PDE-G-CNNs for the Classification of Arteries and Veins</i>	

PRESENTATIONS

- Refereeravond juni 2026**, Refereeravond Oogheelkunde MUMC+ 10 - 06 - 2026
Title: *Het Labyrint van het Netvlies - van Losse Pixels naar een Volledige Vaatboom* (The Labyrinth of the Retina - from Separate Pixels to a Full Vascular Tree)
- REMODEL 2026**, Conference 13 - 05 - 2026
Title: *Connected Components on Lie Groups with Examples in Multi-Orientation Image Analysis*
- Real World Applications of Geometry and Algebra (RAGA) 2025**, Eindhoven University of Technology 27 - 05 - 2025
Title: *Geodesic Tracking of Vascular Trees in Retinal Images*
- CASA Day**, Eindhoven University of Technology 16 - 04 - 2025
Title: *Geodesic Tracking on Retinal Images*
- Ophthalmology Seminar**, Maastricht University 14 - 03 - 2025
Title: *Image Analysis on Retinal Images*
- Algebra Seminar**, Georgia Tech 07 - 10 - 2024
Title: *Lie Groups and Applications to Multi-Orientation Image Analysis*
- Research Talk**, Emory 01 - 10 - 2024
Title: *Lie Groups and Applications to Multi-Orientation Image Analysis*
- Geometry and Machine Learning**, Conference 19 - 06 - 2024
Title: *Geodesic Tracking via Data-Driven Geometry for Vascular Tree Tracking*
- Ophthalmology Seminar**, Maastricht University 31 - 03 - 2023
Title: *Geodesic Tracking of Retinal Vascular Trees with Optical and TV-Flow Enhancement in the Lifted Space of Positions and Orientations $SE(2)$*
- CASA Day**, Evoluon 13 - 04 - 2022
Title: *Geodesic Tracking via New Data-Driven Cartan Connections for Vascular Tree Tracking*

POSTERS

- Crossing-Preserving Geodesic Tracking on Spherical Images**, SSVM 2025 05 - 2025
- Geodesic Tracking of Retinal Vascular Trees with Optical and TV-Flow Enhancement in $SE(2)$** , NMC 2024 04 - 2024
- Geodesic Tracking of Retinal Vascular Trees with Optical and TV-Flow Enhancement in $SE(2)$** , SSVM 2023, CASA day 2024 05 - 2023, 11 - 2023

PUBLICATIONS

- van den Berg, N.J.**, Vis, L., Mula, O., Duits, R. (2026) Connected Components on Lie Groups and Applications to Multi-Orientation Image Analysis. J Math Imaging Vis. <https://doi.org/10.1007/s10851-026-01287-9>
- García-Castellanos, A., Wessels, D.R., **van den Berg, N.J.**, Duits, R., Pelt, D.M., Bekkers, E.J. (accepted to NeurIPS 2025). Equivariant Eikonal Neural Networks: Grid-Free, Scalable Travel-Time Prediction on Homogeneous Spaces. <https://arxiv.org/abs/2505.16035>
- van den Berg, N.J.**, Sherry, F.M., Berendschot, T.T.J.M., Duits, R. (2025). Crossing-Preserving Geodesic Tracking on Spherical Images. In: Bubba, T.A., Gaburro, R., Gazzola, S., Papafitsoros, K., Pereyra, M., Schönlieb, CB. (eds) Scale Space and Variational Methods in Computer Vision. SSVM 2025. Lecture Notes in Computer Science, vol 15668. Springer, Cham. https://doi.org/10.1007/978-3-031-92369-2_15

van den Berg, N.J., Smets, B.M.N., Pai, G., Mirebeau, J.-M., Duits, R. Geodesic Tracking via New Data-Driven Connections of Cartan Type for Vascular Tree Tracking. J Math Imaging Vis (2024). <https://doi.org/10.1007/s10851-023-01170-x>

van den Berg, N.J., Zhang, S., Smets, B.M.N., Berendschot, T.T.J.M., Duits, R. (2023). Geodesic Tracking of Retinal Vascular Trees with Optical and TV-Flow Enhancement in SE(2). In: Calatroni, L., Donatelli, M., Morigi, S., Prato, M., Santacesaria, M. (eds) Scale Space and Variational Methods in Computer Vision. SSVM 2023. Lecture Notes in Computer Science, vol 14009. Springer, Cham. https://doi.org/10.1007/978-3-031-31975-4_40

Kompanets, A., Duits, R., Leonetti, D., **van den Berg, N.**, Snijder, H.H. (2024). Segmentation Tool for Images of Cracks. In: Skatulla, S., Beushausen, H. (eds) Advances in Information Technology in Civil and Building Engineering. ICCCB E 2022. Lecture Notes in Civil Engineering, vol 357. Springer, Cham. https://doi.org/10.1007/978-3-031-35399-4_8

SKILLS

Programming Languages and Frameworks Python (incl. PyTorch), Mathematica, MATLAB, L^AT_EX

Languages Dutch (native), English (fluent), French (intermediate), German (intermediate)